



TEST REPORT

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REPORT No.: CTB25052700602C01

Applicant: Dongguan Kinghal Electronic Technology Co.,Ltd
Address: Room 1301, Building 15, No. 52, Fuhai Road, Xiangang Village, Chang'an Town, Dongguan City, Guangdong Province, China

Manufacturer: Dongguan Kinghal Electronic Technology Co.,Ltd
Address: Room 1301, Building 15, No. 52, Fuhai Road, Xiangang Village, Chang'an Town, Dongguan City, Guangdong Province, China

The following samples were submitted and identified on behalf of the clients as:

Sample name: AI Voice Recorder
Brand: /
Model(s): LA518,LA519,LP039A,L027,L029,LA528,LA537,LA558,LA627,LA636, LA135,LA657
Batch No.: /
Sample received date: May 28, 2025
Testing period: May 28, 2025 to Jun. 05, 2025
Test Method: Please refer to next page(s).
Test Result: Please refer to next page(s).

Result Summary :

Test Requested	Conclusion
European Directive 2011/65/EU and amendment (EU) 2015/863 on the restriction of the use of certain hazardous substances in electrical and electronic equipment	PASS

Tested By: 	Check By: 	Approve By:  
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Date: Jun. 09, 2025

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Test Method:

A. Screening test by XRF spectroscopy

XRF screening limits for regulated elements according to IEC 62321-3-1:2013

Element	Screening limit / mg/kg		MDL	
	Polymers and metals	Composite material	Polymers	Other material
Pb	$BL \leq (700-3\sigma) < X < (1300+3\sigma)$ $\leq OL$	$BL \leq (500-3\sigma) < X < (1500+3\sigma)$ $\leq OL$	10mg/kg	50mg/kg
Cd	$BL \leq (70-3\sigma) < X < (130+3\sigma)$ $\leq OL$	$BL \leq (50-3\sigma) < X < (150+3\sigma)$ $\leq OL$	10mg/kg	50mg/kg
Hg	$BL \leq (700-3\sigma) < X < (1300+3\sigma)$ $\leq OL$	$BL \leq (500-3\sigma) < X < (1500+3\sigma)$ $\leq OL$	10mg/kg	50mg/kg
Cr	$BL \leq (700-3\sigma) < X$	$BL \leq (500-3\sigma) < X$	10mg/kg	50mg/kg
Br	$BL \leq (300-3\sigma) < X$ (non-metal only)	$BL \leq (250-3\sigma) < X$	10mg/kg	50mg/kg

B. Chemical Test

Test Item(s)	Test Method	Analysis Equipment(s)	MDL	Limit
Lead (Pb)	IEC 62321-5:2013	ICP-OES	2 mg/kg	1000mg/kg
Cadmium (Cd)	IEC 62321-5:2013	ICP-OES	2 mg/kg	100mg/kg
Mercury (Hg)	IEC 62321-4:2013+AMD1:2017	ICP-OES	2 mg/kg	1000mg/kg
Hexavalent Chromium Cr(VI)	IEC 62321-7-1:2015	UV-VIS	--	1000mg/kg
	IEC 62321-7-2:2017		8 mg/kg	1000mg/kg
Polybrominated Biphenyls (PBBs)	IEC 62321-6:2015	GC-MS	50mg/kg	1000mg/kg
Polybrominated Diphenyl Ethers (PBDEs)	IEC 62321-6:2015	GC-MS	50mg/kg	1000mg/kg
Phthalate	IEC 62321-8:2017	GC-MS	50mg/kg	1000mg/kg

Tested material list

No.	Description	Photo(s) of tested material
1	Grey metal	
2	Silver metal	
3	Transparent plastic sheet	
4	Black plastic	
5	Yellow plastic film	
6	Black plastic	
7	FPC	
8	Black plastic	
9	Black PCB	
10	Solder point	
11	Grey metal shell	
12	Black PCB	
13	Solder point	
14	Silver metal	
15	Red plastic wire skin	
16	Black plastic wire skin	
17	IC	
18	Crystal oscillator	
19	SMD capacitor	
20	Silver aluminum foil case	

Note: test samples were specified by applicant.

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Test Result(s):

No.	XRF screening Result					Chemical confirm Result (mg/kg)	Remark	Conclusion
	Pb	Cd	Hg	Cr	Br			
1	BL	BL	BL	X	NA	Cr ⁶⁺ : Negative	---	PASS
2	BL	BL	BL	X	NA	Cr ⁶⁺ : Negative	---	PASS
3	BL	BL	BL	BL	BL	---	---	PASS
4	BL	BL	BL	BL	BL	---	---	PASS
5	BL	BL	BL	BL	BL	---	---	PASS
6	BL	BL	BL	BL	BL	---	---	PASS
7	BL	BL	BL	BL	BL	---	---	PASS
8	BL	BL	BL	BL	BL	---	---	PASS
9	BL	BL	BL	BL	X	PBBs: N.D PBDEs: N.D	---	PASS
10	BL	BL	BL	BL	NA	---	---	PASS
11	BL	BL	BL	BL	NA	---	---	PASS
12	BL	BL	BL	BL	X	PBBs: N.D PBDEs: N.D	---	PASS
13	BL	BL	BL	BL	NA	---	---	PASS
14	BL	BL	BL	X	NA	Cr ⁶⁺ : Negative	---	PASS
15	BL	BL	BL	BL	BL	---	---	PASS
16	BL	BL	BL	BL	BL	---	---	PASS
17	BL	BL	BL	BL	BL	---	---	PASS
18	BL	BL	BL	BL	NA	---	---	PASS
19	BL	BL	BL	BL	BL	---	---	PASS
20	BL	BL	BL	BL	NA	---	---	PASS

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Test Item(s)	Dibutyl Phthalate (DBP) (mg/kg)	Benzylbutyl Phthalate (BBP) (mg/kg)	Bis-(2-ethylhexyl) Phthalate (DEHP) (mg/kg)	Diisobutyl phthalate (DIBP) (mg/kg)	Conclusion
CAS No.	84-74-2	85-68-7	117-81-7	84-69-5	
Limit	1000	1000	1000	1000	
No.	Result (mg/kg)				
3+4+5+6	N.D	N.D	N.D	N.D	PASS
1+7+8+9	N.D	N.D	N.D	N.D	PASS
12+17+19	N.D	N.D	N.D	N.D	PASS
15+16	N.D	N.D	N.D	N.D	PASS

Remark:

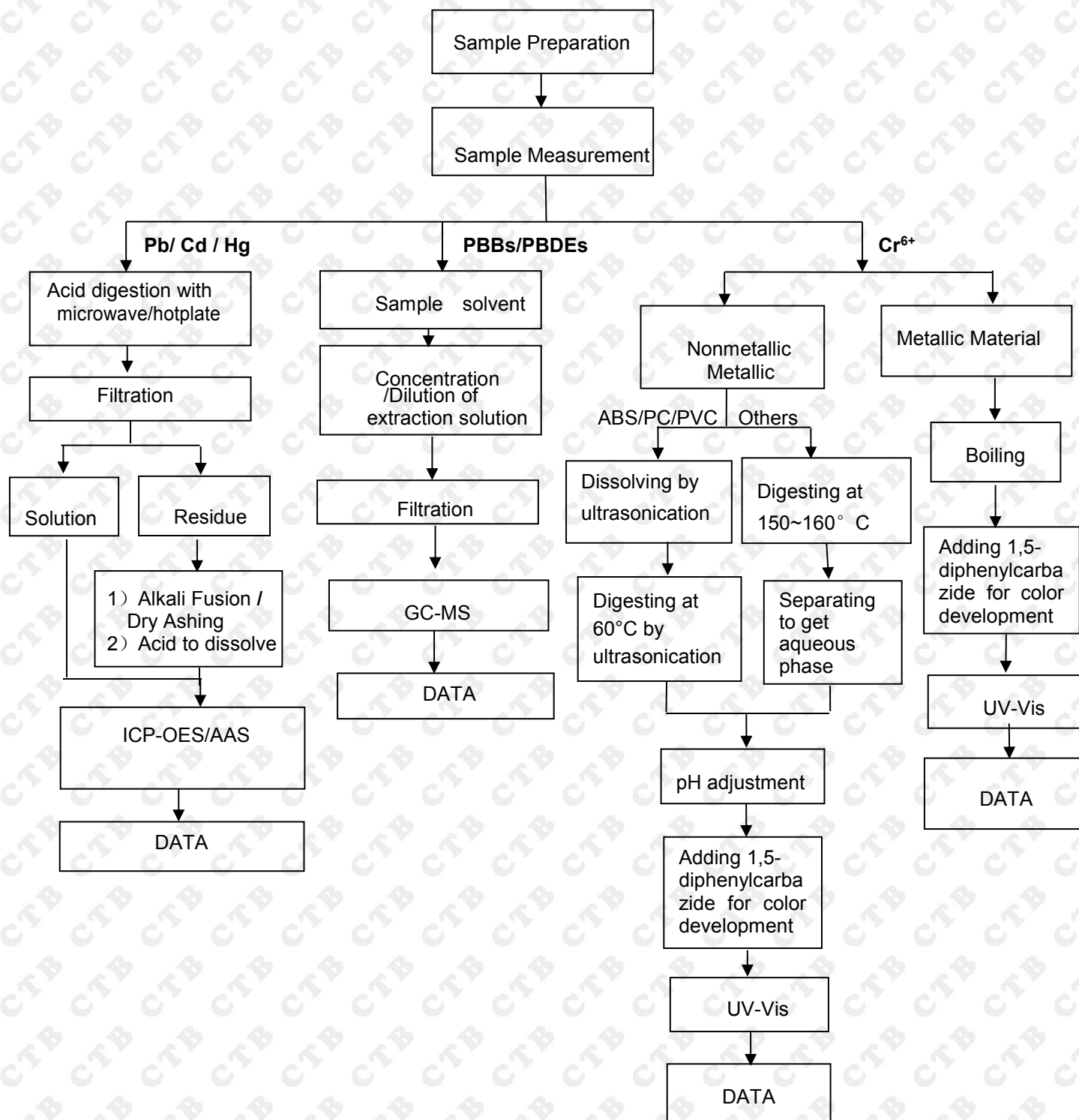
1. BL = below the limit
2. OL = over the limit
3. X = inconclusive, chemical confirm test is needed
4. NA = metal not applicable
5. mg/kg = milligram per kilogram = ppm
6. N.D = not detected
7. Negative = Absence of Cr(VI), the detected Cr(VI) concentration in the boiling water extraction solution is less than $0.1\mu\text{g}/\text{cm}^2$ sample surface area used.
8. Positive = Presence of Cr(VI), the detected Cr(VI) concentration in the boiling water extraction solution is equal to or greater than $0.13\mu\text{g}/\text{cm}^2$
9. sample surface area used. The limit for composite test should be divided by the mixed number.

Note:

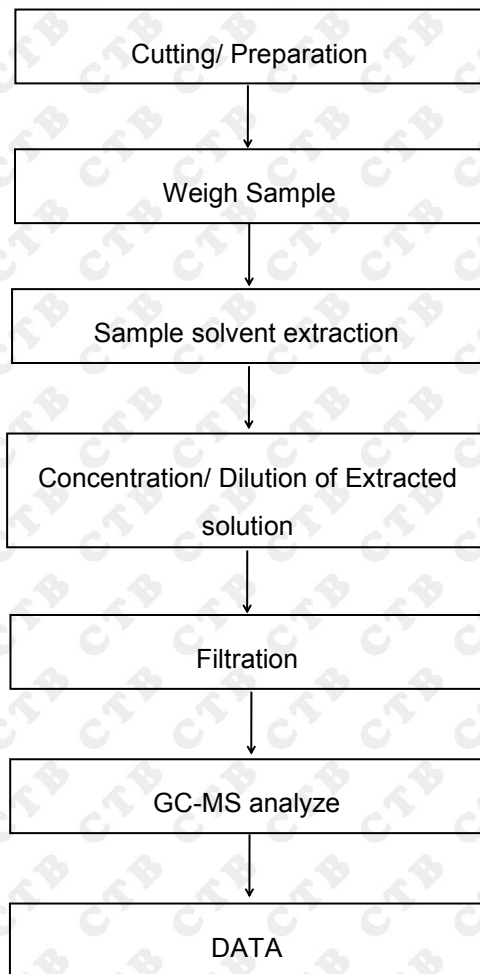
1. When perform screening tests, it is the result on total Br while test item on restricted substances is PBBs/PBDEs, it is the result on total Cr while test item on restricted substances is Cr^{6+} .
2. Pb, Cd, Hg, Cr and Br results are obtained by EDXRF for primary screening, and further chemical testing by ICP-OES (for Cd, Pb, Hg), UV-VIS (for Cr^{6+}) and GC-MS (for PBBs, PBDEs) is needed to be performed, if the concentration falls into the inconclusive area according to IEC 62321-3-1:2013.
3. For the XRF screening test for RoHS elements, the reading may be different to the actual content in the sample be of non-uniformity composition.
4. As requested by the applicant, EU RoHS test was conducted only on components listed in this report. Other components were not tested.

Test flow chart

1. Pb/Cd/Hg/Cr⁶⁺/PBBs/PBDEs



2. Phthalate test flow chart



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Photo documentation



*** End of Report ***

Note: If there is any objection to the inspection results in this report, please submit a written report to the company within 15 days from the date of receiving the report. The test report is effective only with both signature and specialized stamp. This result(s) shown in this report refer only to the sample(s) tested. Without written approval of Shenzhen CTB Testing Technology Co., Ltd. this report can't be reproduced except in full. The tested sample(s) and the sample information are provided by the client. "★" indicates the testing items were fulfilled by subcontracted lab. "※" indicates the items are not in CNAS accreditation scope.

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